

HAT-P-3b

R-1110

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-3_220115 · created 2026-07-30T20:44:20+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2459594.8668 +/- 0.0041 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.135 +/- 0.037
Transit depth (Rp/Rs)^2	1.82 +/- 0.99 [%]
Orbital Inclination (inc)	84.3 +/- 2.41
Airmass coefficient 1 (a1)	0.993 +/- 0.012
Airmass coefficient 2 (a2)	0.0052 +/- 0.0041
Scatter in the residuals of the lightcurve fit is	1.24 %
Best Comparison Star	#2 - [310, 234]
Optimal Aperture	3.9
Optimal Annulus	12.76
Transit Duration (day)	0.038 +/- 0.036

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.135 $\pm$ 0.037 vs 0.1063 (deviation 0.69 σ)
Duration fit vs published	0.038 d vs 0.0867 d (deviation 1.19 σ)
Mid-time O-C	-10.3 min
Depth SNR	3.2
Residual scatter	1.24 %

Light curve

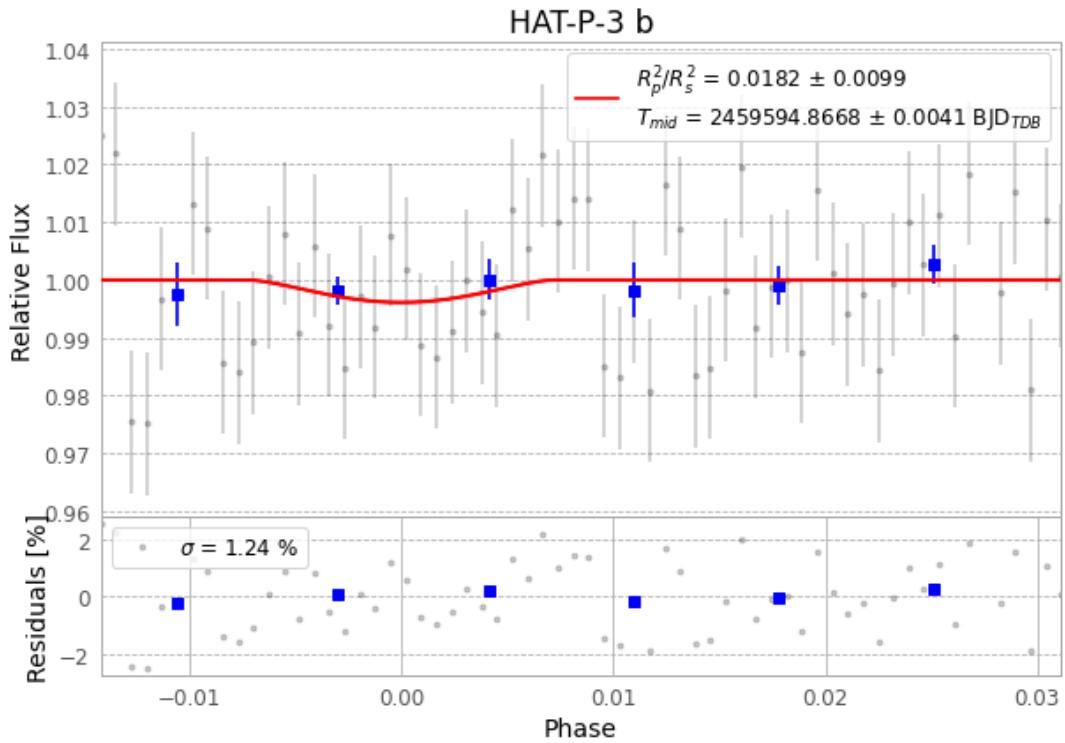


Figure 1 — FinalLightCurve_HAT-P-3 b_2022-01-15.png (SHA-256 a75e28eb45c3a6a4...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [228, 312], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 ed8c0691b5f5af97...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

64 FITS frames (42 MB), HATP-3220115074513.FITS → HATP-3220115105412.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-3-220115.log (f8028ecca1e1...), FinalLightCurve_HAT-P-3 b_2022-01-15.png (a75e28eb45c3...), FinalLightCurve_HAT-P-3 b_2022-01-15.csv (03bad075caa4...)

Compute resources

Wall time 100.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-30 20:44Z.